

Lemma: 设 $(N, 0, s)$ 满足 Peano 公理, 则有: 对 $\forall x \in N$, 有: $x \neq s(x)$

Proof: 对 $\forall x \in N$, 有:

假设 $x = s(x)$, 则有 $x = s(x) \neq 0$.

$\therefore x \in N$ 且 $x \neq 0 \quad \therefore \exists$ 唯一的 $d' \in N$, s.t. $x \neq d'$ 且 $x = s(d')$

$\therefore s(x) = x = s(d')$

$\therefore s: N \rightarrow N$ 是单射 $\therefore x = d'$ 这与 $x \neq d'$ 矛盾

$\therefore x \neq s(x)$



A.2 构造非负整数集

定义 ($\mathbb{Z}_{\geq 0}$ 中的后继元运算) 定义映射 $s: \mathbb{Z}_{\geq 0} \longrightarrow \mathbb{Z}_{\geq 0}$ 为:

$$s: \mathbb{Z}_{\geq 0} \longrightarrow \mathbb{Z}_{\geq 0}$$

$$x \longmapsto x \cup \{x\}.$$

Lemma: 映射 $s: \mathbb{Z}_{\geq 0} \longrightarrow \mathbb{Z}_{\geq 0}$ 是单射, 且对 $\forall x \in \mathbb{Z}_{\geq 0}$, $s(x) \neq 0$.

$$x \longmapsto x \cup \{x\}$$

Proof: 假设 $\exists \zeta \in \mathbb{Z}_{\geq 0}$, s.t. $s(\zeta) = 0$, 则有:

$$\because s(\zeta) = \zeta \cup \{\zeta\}, \quad 0 = \emptyset \quad \therefore \zeta \cup \{\zeta\} = \emptyset$$

$$\therefore \zeta \in \zeta \cup \{\zeta\} = \emptyset \text{ 矛盾.}$$

$$\therefore \text{对 } \forall x \in \mathbb{Z}_{\geq 0}, \text{ 有 } s(x) \neq 0$$

$$\text{对 } \forall x \in \mathbb{Z}_{\geq 0}, \text{ 有: } \because \mathbb{Z}_{\geq 0} \text{ 是归纳集} \quad \therefore x \cup \{x\} \in \mathbb{Z}_{\geq 0}$$

$$\therefore s(x) = x \cup \{x\} \in \mathbb{Z}_{\geq 0} \quad \therefore s(\mathbb{Z}_{\geq 0}) \subseteq \mathbb{Z}_{\geq 0}$$

$$\text{对 } \forall x, y \in \mathbb{Z}_{\geq 0}, \text{ 有:}$$

$$\text{若 } x = y, \text{ 则有 } \{x\} = \{y\}, \quad \therefore s(x) = x \cup \{x\} = y \cup \{y\} = s(y)$$

$$\therefore s: \mathbb{Z}_{\geq 0} \longrightarrow \mathbb{Z}_{\geq 0} \text{ 是一个映射.}$$

$$\text{若 } s(x) = s(y), \text{ 则有: } x \cup \{x\} = y \cup \{y\} \quad \therefore x \in \{x\} \subseteq x \cup \{x\} = y \cup \{y\}$$

$$\therefore x \in y \text{ 或 } x \in \{y\}.$$

$$(i) \text{ 若 } x \in \{y\}, \text{ 则 } x = y.$$

$$(ii) \text{ 若 } x \in y, \text{ 则: } \because y \in \mathbb{Z}_{\geq 0} \quad \therefore y \text{ 是传递集} \quad \therefore x \subseteq y.$$

$$\therefore y \in y \cup \{y\} = x \cup \{x\}$$

$$\therefore y \in x \text{ 或 } y \in \{x\}$$

$$\text{如果 } y \in \{x\}, \text{ 则直接得到 } y = x \quad \therefore x = y$$

$$\text{如果 } y \in x, \text{ 则 } \therefore x \in \mathbb{Z}_{\geq 0} \quad \therefore x \text{ 是传递集} \quad \therefore y \subseteq x$$

$$\therefore x \subseteq y \text{ 且 } y \subseteq x \quad \therefore x = y$$

$$\therefore s: \mathbb{Z}_{\geq 0} \longrightarrow \mathbb{Z}_{\geq 0} \text{ 是单射.}$$



定理: 资料 $(\mathbb{Z}_{\geq 0}, 0, s)$ 满足 Peano 公理.

Proof: 零非后继已证, 后继无运算是单射已证.

设 $I \subseteq \mathbb{Z}_{\geq 0}$ 为包含 0 的子集, 而且对 $\forall x \in \mathbb{Z}_{\geq 0}$ 都有 $x \in I \Rightarrow s(x) \in I$,

则有: $\therefore I \subseteq \mathbb{Z}_{\geq 0} \quad \therefore I$ 是一个集合.

$$\therefore 0 \in I \quad \therefore \emptyset = 0 \in I$$

$$\text{若 } \alpha \in I, \text{ 则有: } \alpha \in I \subseteq \mathbb{Z}_{\geq 0} \quad \therefore s(\alpha) \in I$$

$$\therefore \alpha \cup \{\alpha\} \in I$$

$\therefore I$ 是归纳集.

$\therefore \mathbb{Z}_{\geq 0}$ 是最小的归纳集

$$\therefore \mathbb{Z}_{\geq 0} \subseteq I$$

$$\therefore I \subseteq \mathbb{Z}_{\geq 0} \text{ 且 } \mathbb{Z}_{\geq 0} \subseteq I$$

$$\therefore I = \mathbb{Z}_{\geq 0}$$

$\therefore$ 数学归纳法原理得证.



Lemma: $\nexists \forall x, y \in \mathbb{Z}_{\geq 0}$, 若 $x \in y$, 则有: $x \subseteq y$.

Proof: $\because y \in \mathbb{Z}_{\geq 0}$

$\therefore y$ 是传递集.

$\because x \in y$

$\therefore x \subseteq y$



~~Lemma: $\nexists \forall x, y \in \mathbb{Z}_{\geq 0}$, 若 $x \in y$, 则有: $x \leftarrow y$.~~

~~Proof:~~

~~Lemma: $\nexists \forall x, y \in \mathbb{Z}_{\geq 0}$ $\equiv$~~

Lemma: $\nexists \forall x \in \mathbb{Z}_{\geq 0}$, 有: $x \in S(x)$

Proof: $\nexists \forall x \in \mathbb{Z}_{\geq 0}$, 有: $S(x) \in \mathbb{Z}_{\geq 0}$

$\because S(x) = x \cup \{x\}$

$\therefore x \in \{x\} \subseteq x \cup \{x\} = S(x)$



Lemma: 对 $\forall x, y \in \mathbb{Z}_{\geq 0}$, 若 $x \in y$, 则有: 对 $\forall \lambda \in \mathbb{Z}_{\geq 0}$, 有:
 $x \in (y + \lambda)$

Proof: 令 $I = \{\lambda \in \mathbb{Z}_{\geq 0} \mid x \in (y + \lambda)\}$, 则有: $I \subseteq \mathbb{Z}_{\geq 0}$

$$\because \cancel{y} \in \mathbb{Z}_{\geq 0} \quad \therefore y + 0 = y \quad \therefore x \in y = y + 0$$

$\therefore 0 \in \mathbb{Z}_{\geq 0}$ 且 $x \in (y + 0) \quad \therefore 0 \in I \quad \therefore I$ 是 $\mathbb{Z}_{\geq 0}$ 的包含 0 的子集.

对 $\forall \lambda \in \mathbb{Z}_{\geq 0}$, 若 $\lambda \in I$, 则有:

$$\therefore \lambda \in \mathbb{Z}_{\geq 0} \quad \therefore s(\lambda) \in \mathbb{Z}_{\geq 0}$$

$$\therefore y \in \mathbb{Z}_{\geq 0} \text{ 且 } s(\lambda) \in \mathbb{Z}_{\geq 0} \quad \equiv$$

$$\therefore y + s(\lambda) = s(y + \lambda) = (y + \lambda) \cup \{y + \lambda\}$$

$$\therefore \lambda \in I \quad \therefore x \in (y + \lambda) \quad \therefore x \in (y + \lambda) \subseteq (y + \lambda) \cup \{y + \lambda\} = y + s(\lambda)$$

$$\therefore s(\lambda) \in I$$

$$\therefore I = \mathbb{Z}_{\geq 0}$$

$$\therefore \text{对 } \forall \lambda \in \mathbb{Z}_{\geq 0}, \text{ 有: } \lambda \in \mathbb{Z}_{\geq 0} = I \quad \therefore x \in (y + \lambda) \quad \square$$

Lemma: 对 $\forall x, y \in \mathbb{Z}_{\geq 0}$, 若 $x < y$, 则有: $x \in y$.

Proof: $\because x < y \quad \therefore x \leq y$ 且 $x \neq y$

$\because x \leq y \quad \therefore \exists \alpha \in \mathbb{Z}_{\geq 0}$, s.t. $y = x + \alpha$

假设 $\alpha = 0$, 则有: $y = x + \alpha = x + 0 = x \quad \therefore x = y$. 矛盾.

$\therefore \alpha \neq 0$.

$\because \alpha \in \mathbb{Z}_{\geq 0}$ 且 $\alpha \neq 0 \quad \therefore \exists$ 唯一的 $d' \in \mathbb{Z}_{\geq 0}$ 使得 $\alpha \neq d'$ 且 $\alpha = s(d')$

$\therefore y = x + \alpha = x + s(d') = s(x + d') = s(x) + d'$

$\because x \in \mathbb{Z}_{\geq 0} \quad \therefore x \in s(x)$

$\because x \in \mathbb{Z}_{\geq 0}, s(x) \in \mathbb{Z}_{\geq 0}, x \in s(x), d' \in \mathbb{Z}_{\geq 0}$

$\therefore x \in (s(x) + d') = y \quad \therefore x \in y \quad \square$

Lemma: 对 $\forall x, y \in \mathbb{Z}_{\geq 0}$, 若 $x \in y$, 则有: $x < y$

Proof: 假设 $x \geq y$, 则有: $y \leq x \quad \therefore y < x$ 或 $y = x$.

若 $y < x$, 则有 $y \in x \quad \therefore x \in y \quad \therefore x \leq y \quad \therefore y \in x \leq y$

$\therefore y \in y$.

若 $y = x$, 则 $\therefore x \in y \quad \therefore \text{~~x \in y~~ } y \in y$.

$\equiv \therefore y \in y \quad \therefore \{y\} \subseteq y \quad \therefore y \cup \{y\} = y \quad \therefore s(y) = y \cup \{y\} = y$.

矛盾. $\therefore x < y. \quad \square$

Lemma: $\forall x, y \in \mathbb{Z}_{\geq 0}$, 有:

$$x < y \iff x \in y$$

Proof: 由上面两个引理立得. $\square$